SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 5 **Duration : 1 Hour**
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I __J.Jeshwanth(1921260264)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

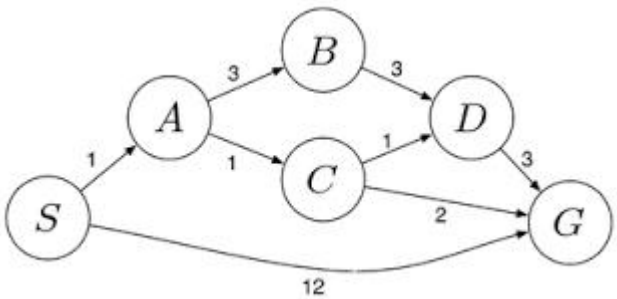
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

   The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

1. Water Jug Problem

Problem: There are two jugs of capacities 4 gallons and 3 gallons. The objective is to obtain exactly 2 gallons in the 4-gallon jug without using any measuring device.

Solution steps:

Step	Action	4-gallon jug	3-gallon jug
1	Fill the 3-gallon jug	0	3
2	Pour into the 4-gallon jug	3	0
3	Fill the 3-gallon jug again	3	3
4	Pour into the 4-gallon jug until full	4	2
5	Empty the 4-gallon jug	0	2
6	Pour the remaining water into the 4-gallon jug	2	0

Final answer: Exactly 2 gallons of water are obtained in the 4-gallon jug.

2. Mars Rover as an Intelligent Agent

A Mars Rover is an intelligent agent that explores the surface of Mars, collects samples, and analyses the environment to support scientific research.

i. What are the percepts of this agent?

The percepts are the information received through its sensors:

Images of the Martian surface.

Temperature, pressure, and environmental conditions.

Rocks, soil, and mineral composition.

Obstacles, slopes, and terrain conditions.

Distance and position information.

ii. Characterize the operating environment.

The Mars Rover operates in an environment with the following properties:

Property	Characteristic

Observability	Partially observable
Determinism	Stochastic
Episodic or sequential	Sequential
Static or dynamic	Dynamic
Discrete or continuous	Continuous
Number of agents	Single-agent

iii. What actions can the agent take?

Move forward, backward, and turn.

Capture images using cameras.

Collect and analyse soil and rock samples.

Drill rocks and use scientific instruments.

Communicate collected data to Earth.

Avoid obstacles and navigate around hazards.

iv. How can we evaluate the performance of the agent?

Performance can be evaluated based on:

Number and quality of samples collected.

Accuracy of scientific analysis.

Distance explored.

Successful completion of assigned missions.

Energy consumption and equipment safety.

Quality and quantity of useful data transmitted to Earth.

v. What is the most suitable agent architecture? Why?

A model-based, goal-based intelligent agent is suitable for the Mars Rover.

Reason: It maintains an internal model of its environment despite incomplete sensor information and selects actions to achieve goals such as exploring terrain,

collecting samples, and avoiding obstacles. It can also plan routes and adapt to changing conditions.

3. Eight Queens Problem

Problem: Place eight queens on an \(8 \times 8\) chessboard so that no two queens attack each other.

Problem formulation

Initial state: An empty chessboard.

State: A configuration containing zero or more queens placed on the board.

Actions: Place a queen in a safe square, one column at a time.

Transition model: The board changes by adding a queen to the selected square

Goal test: Eight queens are placed, with no two sharing the same row, column, or diagonal.

Path cost: Each queen placement has a cost of 1; a complete solution costs 8.

Search strategy

Backtracking search can be used to solve the problem.

Start with the first column.

Place a queen in a safe row.

Move to the next column and place another safe queen.

If no safe square is available, remove the previous queen and try another row.

Repeat until all eight queens are placed.

One valid solution, represented by the row number selected in each column, is:

\[[1,5,8,6,3,7,2,4] \]

Each queen occupies a different row and column, and no two queens share a diagonal.

4. OLA Cab Booking Application

Problem: A customer wants to travel from a starting location to a destination using the OLA application and select a suitable cab type.

Type of problem-solving agent

A goal-based agent is suitable because it selects actions to achieve the goal of reaching the destination. A utility-based component can also compare cab options based on cost, comfort, and travel time.

Pseudocode

```
BEGIN
   Get customer's starting location
   Get customer's destination
   Display available cab types
   Get customer's preferred cab type
   Find available cabs
   Calculate estimated fare and travel time

   IF cab is available THEN
      Display fare and arrival time
      Confirm booking
      Assign cab to customer
      Track cab until destination
   ELSE
      Display "Cab not available"
   END IF
END
```

Result: The agent helps the customer select and book a suitable cab to reach the destination.

5. Uniform Cost Search (UCS)

Problem: Find the minimum-cost path from the starting warehouse \(S\) to the destination warehouse \(G\) in the given weighted graph.

Algorithm

Uniform Cost Search expands the node with the lowest cumulative path cost first. It uses a priority queue ordered by total cost \(g(n)\).

Step-by-step calculation

From the graph given in the question, the edges and costs are:

Route	Cost
S → A	1
S → G	12
A → B	3
A → C	1
B → D	3
C → D	1
C → G	2
D → G	3

The UCS search proceeds as follows:

Step	Node expanded	Path cost	Frontier after expansion (node: cost)
1	S	0	A: 1, G: 12
2	A	1	C: 2, B: 4, G: 12
3	C	2	D: 3, B: 4, G: 4
4	D	3	B: 4, G: 4
5	B	4	G: 4
6	G	4	Goal reached

At the end, \(G\) is removed from the priority queue with the lowest cost of 4. The path through \(S \rightarrow A \rightarrow C \rightarrow G\) is the optimal solution.

Pseudocode

```
BEGIN
   Insert S with cost 0 into priority queue

   WHILE priority queue is not empty
      Remove node with minimum cost

      IF node = G
         Return path and cost
```

```
        FOR each neighbour of node
            Calculate new path cost
            IF new path is cheaper
                Update priority queue
            END IF
        END FOR
    END WHILE
END
```

Final answer

The optimal path is:

$$ S \rightarrow A \rightarrow C \rightarrow G $$

Total cost:

$$ 1+1+2=\boxed{4} $$

Result: Uniform Cost Search finds the least-cost delivery route with a total cost of 4 units.